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SPECIFIC HEAT EVIDENCE ON THE NORMAL STATE PSEUDOGAP

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Abstract—The electronic specific heat of underdoped cuprates reveals an absence of spectral weight below a shoulder at the pseudogap energy E_g. With hole doping additional spectral weight with entropy $\sim k_B$/added hole appears on the flanks of the gap and E_g decreases as $E_g = J(1 - p/p_{crit})$, where $J \approx 1300$ K and $p_{crit} \sim 0.18/CuO_2$ plane. This leads to an abrupt cross-over to metallic behaviour and strong superconductivity when the gap closes at $p_{crit} \sim J\gamma_n/k_B$. We conclude that $E_g(p)$ reflects the energy scale of correlated holes and spins, and is not due to superconducting fluctuations or additional competing interactions. © 1998 Elsevier Science Ltd. All rights reserved

1. INTRODUCTION

In this paper we review evidence from the electronic specific heat $\gamma(T) = C^{el}/T$ on the normal state (NS) pseudogap and its influence on the superconducting (SC) condensate. A widely held view supported by ARPES measurements on underdoped BSCCO [1, 2] is that the pseudogap reflects precursor SC correlations without long range phase coherence, so that the pseudogap energy scale E_g is simply a continuation to above T_c of the SC gap $\Delta(T)$ [3]. We test this hypothesis with our data for $Y_{0.8}Ca_{0.2}Ba_2Cu_3O_{7-\delta}$ [4] and find that it cannot be sustained. We then show that this data and data for $La_{2-x}Sr_xCuO_4$ [5] provide a simple alternative view of E_g as the characteristic energy of correlated holes and spins.

The electronic entropy $S(T) = \int^T \gamma(T')dT'$ gives the integrated spectral weight within an energy range $\sim E_F \pm 2T$, whilst $\gamma(T)$ vs T reflects the energy dependence of the DOS g(E) vs $|E - E_F|$ (thermally broadened and with approximate scaling $|E - E_F| \sim 2 - 3T$). The initial linear T-dependences of γ and χ provided the first evidence for a linear energy dependence $g(E) \sim |E - E_F|/E_g$ for the pseudogap DOS [6], an early indication of the anisotropic angular dependence later confirmed by ARPES [1]. Either triangular-like (states-non-conserving) or d-wave-like (states-conserving) gaps of half-width E_g yield this initial linear T-dependence and predict a knee at around $0.4E_g$ and $0.6E_g$ in γ and S/T respectively. Distinguishing between these two forms of the gap DOS from γ vs T is inconclusive since the underlying DOS may be energy dependent (see LSCO below). However a crucial difference is that entropy lost at low T is ultimately regained at sufficiently high T if the gap DOS conserves states, but is permanently lost for any states-non-conserving gap ($\ln(4)k_B$/state for conventional fermions). Similar conclusions apply to $\chi T \propto < \mu_z^2 >$ where χ is the spin susceptibility [6]. The observed permanent loss of entropy for underdoped cuprates (see below) strongly supports a states-non-conserving pseudogap.

2. EVIDENCE FROM $Y_{0.8}Ca_{0.2}Ba_2Cu_3O_{7-\delta}$.

The overdoped region of $YBCO_{7-\delta}$ is extended by Ca doping, and $Y_{O.8}Ca_{O.2}Ba_2Cu_3O_{7-\delta}$ with $\delta_{opt} \sim 0.32$ is an ideal system for the investigation of SC and NS gaps [4]. γ, S and S/T are shown in Fig. 1(a) for the overdoped region and Fig. 1(b) for the underdoped region to emphasize the sudden change of behaviour at around $\delta_{crit} \sim 0.25$. In overdoped samples the behaviour is typical of a conventional d-wave superconducting metal with T_c and $\Delta(0)$ increasing with Δ [4], γ_n independent of δ, and most importantly, no evidence for a loss of entropy above T_c from a NS pseudogap. This behaviour changes abruptly at around δ_{crit}, with a sudden collapse of the specific heat jump (a very sensitive indicator of the initial appearance of the pseudogap) and a more or less T-independent loss of entropy above T_c yielding the characteristic pseudogap T-dependence for S/T. The behaviour in this region is very similar to that of $YBCO_{7-\delta}$ for $\delta > 0.05$ [7] showing little further change with δ in the spectral gap below T_c.

Fig. 2 shows that above T_c a T-independent pseudogap DOS provides a good fit to S/T and γ for a typical underdoped sample $\delta = 0.439$. We see no evidence for a phase transition or onset temperature T^* for the pseudogap which would be revealed by a kink in S/T and a jump in $\gamma = dS/dT$ at T^*. We cannot exclude the possibility of a weaker T-dependence for the NS DOS but the permanent loss of high temperature entropy [Fig. 1(b)] shows that this can at most be a redistribution rather than a restoration of lost spectral weight.

The equivalent d-wave energy gap $\Delta(T)$ that yields the measured entropy at each T [7], is shown in Fig. 3. Each curve shows a kink at T_c, reflecting the kink in S/T and

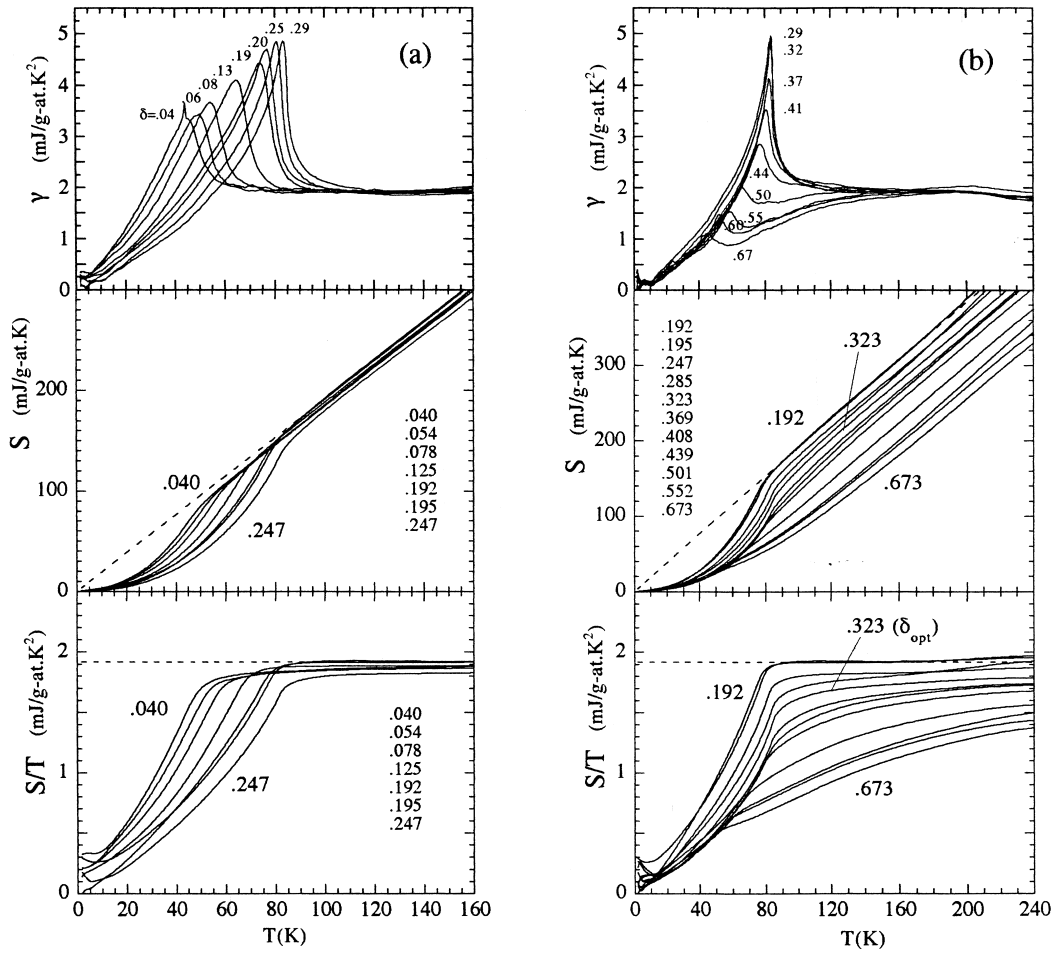


Fig. 1. γ, entropy S and S/T for (a) overdoped and; (b) underdoped $Y_{0.8}Ca_{0.2}Ba_2Cu_3O_{7-\delta}$. Labels show δ. Note the abrupt onset of the pseudogap at $\delta_{crit} \sim 0.25$.

jump in γ. Below T_c, $\Delta(T)$ has a BCS-like T-dependence reflecting cooperative SC order whilst above T_c, $\Delta(\equiv E_g)$ is essentially T-independent reflecting an energy scale for incoherent correlations. Note that on a fluctuation interpretation of E_g the SC order parameter below T_c is the spectral gap $\Delta(T)$, but if SC and the pseudogap are independent the order parameter is $\sim\Delta(T) - E_g$, and is suppressed to $T = 0$ [7].

Fig. 2 illustrates our procedure for estimating the SC condensation energy $U(0)$ in the pseudogap region, where

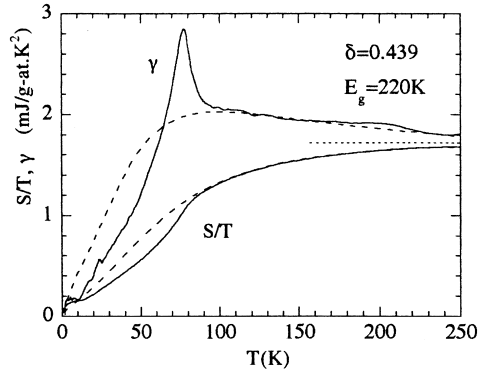


Fig. 2. Broken lines show fits of γ and S/T to a T-independent d-wave-like pseudogap for YBCO(20% Ca) with $\delta = 0.439$. Note the strongly reduced condensation energy, related to the area between $(S/T)_n$ and $(S/T)_s$.

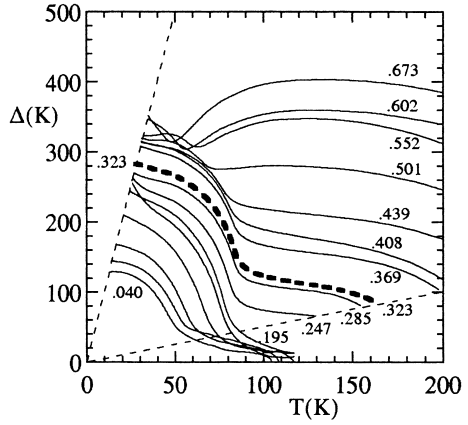


Fig. 3. Equivalent d-wave energy gap $\Delta(T)$ ($\equiv E_g$ above T_c) for YBCO(20% Ca) deduced from S/T.

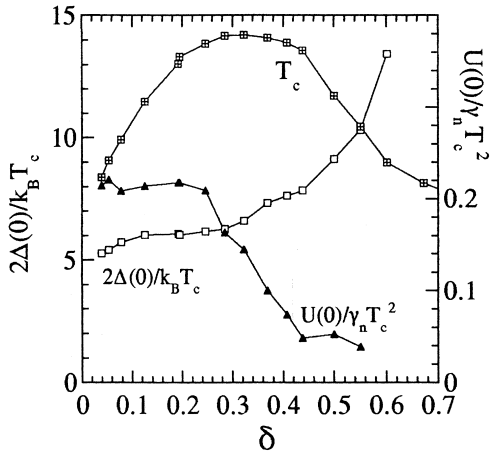


Fig. 4. $U(0)/\gamma_n T_c^2$ and $2\Delta(0)/k_B T_c$ for YBCO(20% Ca), where $U(0)$ is the condensation energy.

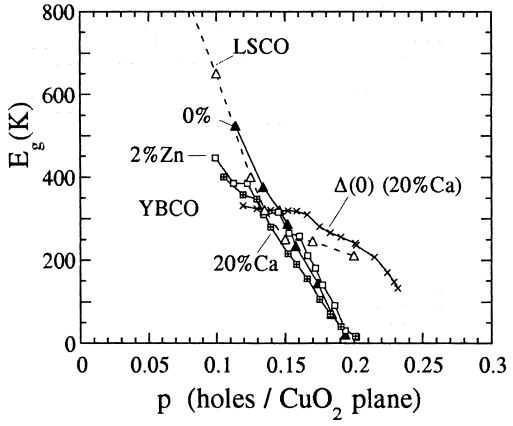


Fig. 6. E_g vs planar hole density p for $La_{2-x}Sr_xCuO_4$ and $YBCO_{7-\delta}$(O, 2% Zn and 20% Ca). $\Delta(0)$ for YBCO(20% Ca) is shown by crosses.

$U(0) = \int^{T_c}(S_n - S_s)dT'$. We take $S_s(T')$ from experiment and $S_n(T')$ from a continuation of the predicted normal state entropy curve to below T_c (e.g. from the broken curve for $(S/T)_n$ in Fig. 2). The strong reduction in $U(0)$ accompanying the opening of the pseudogap is obvious from the strongly reduced area between $(S/T)_n$ and $(S/T)_s$ shown in Fig. 2 for $\delta = 0.439$ (and indeed for any of the underdoped samples) compared with the corresponding areas for overdoped samples (Fig. 1). $U(0)$ vs δ has been shown previously [4] and is sharply peaked around δ_{crit}. As shown in Fig. 4, $U(0)/\gamma_n T_c^2$ and $2\Delta(O)/k_B T$ are constant for $\delta < \delta_{crit}$ but show anomalous behaviour for $\delta > \delta_{crit}$, further illustrating the abrupt change from conventional metallic to pseudogap behaviour at δ_{crit}.

Note that our assumed continuation of $S_n(T')$ to below T_c is only appropriate if the pseudogap exists in the NS below T_c. This would not be correct if the entropy loss above T_c were due to SC fluctuations, when a better choice would be $(S/T)_n \sim \gamma(300\ K)$ (Fig. 1). This leads to much larger values for $U(0) \sim \gamma(300\ K)\Delta(0)^2$ and no suppression of $U(0)$ in the underdoped region. This dilemma, related to the choice between $\Delta(T) - E_g$ or $\Delta(T)$ for the order parameter below T_c, cannot be resolved from the specific heat or other measurements of the spectral gap. Instead we must compare the contrasting predictions for $U(0)$ with other measures of the strength of the condensate. The superfluid density, diamagnetism and irreversibility field are all found to fall sharply for $\delta > \delta_{crit}$ in YBCO(20%Ca) and indeed in the underdoped region of all cuprates. This confirms that the pseudogap (loss of low energy NS spectral weight) persists to $T \ll T_c$, suppressing $U(0)$ and weakening the condensate, and is strong evidence against a precursor fluctuation interpretation of the pseudogap above T_c.

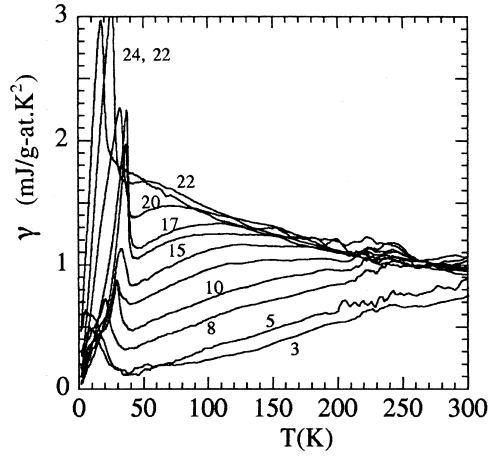


Fig. 5. γ vs T for $La_{2-x}Sr_xCuO_4$. Labels show x% Sr.

Other aspects of our results are inconsistent with precursor fluctuation models. (i) These models predict a gradual p-dependence of $E_g \propto T_c^{MF}$ [3] rather than the observed abrupt onset at $p_{crit} \sim p_{opt}$. The absence of entropy loss and p-independent values of $2\Delta(0)/k_B T_c$ and $U(0)/\gamma_n T_c^2$ confirm the absence of the NS pseudogap for $p > p_{crit}$. (ii) The entropy loss for $p < p_{crit}$ is not restored up to room temperature (4–$6T_c$). (iii) $\Delta(0)$ and E_g exhibit very different doping dependences, shown against δ in Ref.[4] and against p in Fig. 6. (iv) Zn doping reduces the SC gap $\Delta(0)$ of optimally doped $YBC0_{7-\delta}$ [8] but does not affect E_g or the high temperature entropy loss for $p < p_{crit}$ (Figs 6 and 7). This shows that E_g is not related to T_c^{MF}.

3. AN ALTERNATIVE DESCRIPTION OF THE PSEUDOGAP

Unlike the states-conserving SC behaviour of overdoped samples [Fig. 1(a)] the permanent entropy loss due to the pseudogap in the underdoped region cannot be readily explained in terms of a metallic DOS with states removed from E_F and redistributed around E_g. The behaviour in

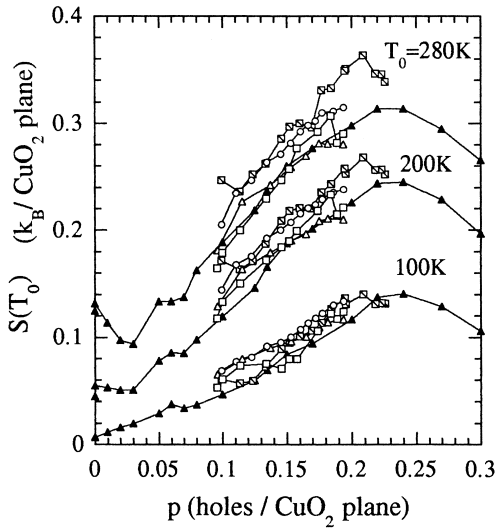


Fig. 7. Entropy $S(T_0)$ at fixed $T = T_0$ vs p, reflecting spectral weight within fixed energy windows $\sim E_F \pm 2k_BT_0$. Closed triangles, $La_{2-x}Sr_xCuO_4$; open symbols, $YBCO_{7-\delta}$(0, 2, 7% Zn and 20% Ca).

this region is better described as an evolution with hole doping from the spectrum of collective spin excitations of the parent AF insulator (for which $\gamma \propto T$ for $T \ll J$). This is illustrated both in Fig. 1(b) and in Fig. 5 showing $\gamma(T)$ for $La_{2-x}Sr_xCuO_4$ [5]. Whilst the T-dependences for LSCO differ markedly from YBCO based materials [4, 8], a gap-like T-dependence moving to lower temperatures with increased doping is observed up to $x\sim0.24$ with energy scale $E_g\sim1550(1 - x/0.17)$ (K) for $x \leq 0.15$ [9]. As for $YBCO_{7-\delta}$ and YBCO(20% Ca), entropy plots [5] show parallel shifts $\sim k_B$/hole with doping, due to added spectral weight around the flanks of the gap (evident from the added areas under $\gamma(T)$ in Fig. 5).

Two unifying properties of the YBCO and LSCO systems are illustrated in Figs 6 and 7. When plotted against the hole density p/CuO_2 plane [taking $p = x$ for LSCO, $0.2(1 - \delta)$ for YBCO(0,2% Zn) and $0.2(1 - +$ 0.2) for YBCO(20% Ca)] E_g is almost identical for each system (Fig. 6) and given by $E_g\sim J(1 - p/p_{crit})$ with $J\sim1200 - 1500$ K and $p_{crit}\sim0.17 - 0.19$. The proximity of J to the in-plane exchange energy $J\sim$O.13 eV has been noted previously [10] and clearly suggests a magnetic origin for the pseudogap. Fig. 7 shows that the entropy at fixed temperature $S(T_O)$ (ie the spectral weight within a fixed energy window $\sim E_F \pm 2T_O$) increases linearly with p at the same rate for each system, with slope 1.3 k_B/hole at $T_O = 280$ K close to the maximum value for fermions of $\ln(4)k_B$/ orbital [6]. (Note that for a metal, $S(T_O) = \gamma_n T_O$ would be independent of p). We have found for YBCO and LSCO a similar linear dependence of $\chi T(T_O)$ vs p with slope close to the value $0.5\mu_B^2$/hole expected for fermions [5, 6], confirming that the the pseudogap at E_g is related to the energy distribution of (damped) quasiparticles and not simply to AF correlations. This identity between the extra low energy weight as the pseudogap closes and the chemical doping exposes an intimate relationship between the pseudogap and the semiconductor-like properties of underdoped cuprates and shows that the latter are not inconsistent with a large Fermi volume.

As noted above the extra weight appears at $E_g\sim J$ for low p and at progressively lower energies as p increases (around the flanks of the gap at $E_F \pm E_g$). From the added entropy $S\sim pk_B$ the gap must close more or less abruptly at $p_{crit}\sim J\gamma_n/k_B\sim Jg(E)$ where γ_n and $g(E)$ are mean values per CuO_2 plane. In terms of the kinetic energy of non-interacting holes $E_M(p)\sim pk_B/\gamma_n$ we have $E_M(p_{crit})\sim J$ and $E_g\sim J(1 - p/p_{crit})\sim J - E_M(p)$. We conclude that the energy of correlated holes and spins softens towards the chemical potential as p approaches p_{crit}, allowing strong superconductivity to develop only when sufficient low energy spectral weight is available. A crossover from semiconducting to metallic behaviour is expected when p exceeds p_{crit}, ie when $E_M(p) > J$, but since from experiment $p_{opt}\sim p_{crit}$ it is evident that the optimum conditions for superconducting pairing occur when $E_M(p)\sim J$. It is clear that the pseudogap, through the three apparently universal parameters J, P_{crit} and spectral weight per doped hole, controls both the superconducting and normal state properties of underdoped cuprates.

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